7 **Ans: B**

Vertical motion (Time of flight)

Initial vertical speed is zero. The vertical distance (y) it falls is determined by:

$$y = \tfrac{1}{2} gt^2$$

Since the distance (y) is directly proportional to the square of the time (t^2), multiplying the distance by 4 means we multiply the time by the square root of 4. Therefore, the second object is in the air for exactly twice as long (2t).

Horizontal motion (Range)

Horizontal distance (x) is not affected by gravity, so for constant speed: $x = vt$

For the first object: $x_1 = vt$

For the second object: $x_2 = v(2t) = 2x$

8 **Ans: D**